

SOFTWARE ENGINEERING

Lab 1: Requirements Engineering & UML Use-Case Modelling

Problem Statement #03 – Campus & Academic Operations Campus Placement & Internship Pipeline

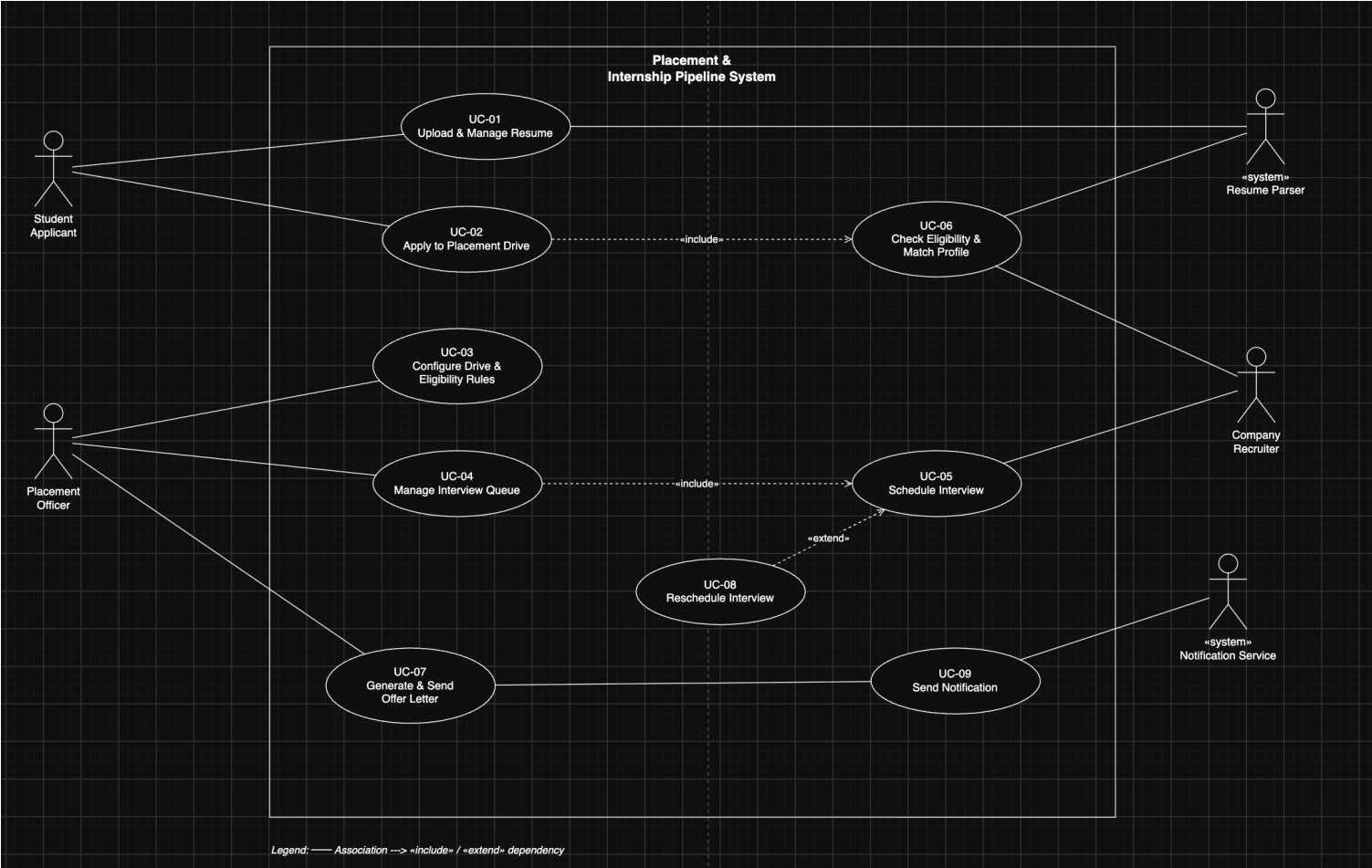
Name: Shreyas K M
SRN: PES1UG24AM402
SEC: G

1. Requirements Table

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-001	Functional	The system shall parse uploaded student resumes (PDF format), extract skill tags, and match profiles against active company eligibility rules.	High	Pass: Ineligible applicants are filtered out with clear discrepancy reasons. Fail: Applicant with an active backlog bypasses the eligibility filter.	Automated screening is the entry point of the pipeline; without accurate parsing and matching, every downstream stage (interviews, offers) operates on unreliable data.
FR-002	Functional	The system shall allow a Placement Officer to configure a corporate drive with eligibility rules, including minimum CGPA, maximum permitted active backlogs, and eligible branches.	High	Pass: A drive saved without all three eligibility fields defined is rejected with a validation error. Fail: A drive is published while one or more eligibility fields are blank.	Companies specify different eligibility tiers per drive; configurable rules are required before FR-001 has anything to match against.
FR-003	Functional	The system shall allow an eligible Student Applicant to submit an application to a published drive and shall queue the application for Round 1 of the interview process.	High	Pass: After a successful application, the student's dashboard shows status "Applied - Round 1 Queued" within 5 seconds. Fail: An ineligible student is able to submit an application.	This is the core self-service action that converts an eligible profile into an active pipeline entry.

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-004	Functional	The system shall allow the Placement Officer or Company Recruiter to advance an applicant through multiple interview rounds, recording a result (Selected / Rejected / On-Hold) after each round.	Medium	Pass: Recording a "Selected" result for Round N automatically updates the applicant's status to "Round N+1 Queued" (or "Final Selection" if it is the last round). Fail: Applicant status remains unchanged after a round result is recorded.	Reflects the multi-round interview queue described in the problem statement and keeps students and staff informed of live status.
FR-005	Functional	The system shall auto-generate an offer letter for a Student Applicant marked "Selected" in the final round, and shall notify the student by email with an option to accept or decline within a configurable deadline.	High	Pass: A student reaching final selection receives an offer email with accept/decline actions within 10 minutes, and the acceptance/decline is time-stamped on record. Fail: No offer letter or notification is generated after final selection is confirmed.	Automates and creates an auditable record of the pipeline's most consequential and legally relevant step.
NFR-001	Nonfunctional	The system shall encrypt all student academic records and offer documents at rest using AES-256.	High	Pass: Benchmarking tests confirm target latency and security standards under simulated peak load. Fail: Any academic record or offer document is found stored unencrypted at rest.	Academic records and offer letters are sensitive personal data; encryption at rest is required to prevent data breaches and meet institutional data-protection policy.
NFR-002	Nonfunctional	The system shall support at least 2,000 concurrent Student Applicant users during peak application windows (e.g., drive registration deadlines) with a 95th-percentile response time of no more than 3 seconds per request.	High	Pass: Load testing with 2,000 concurrent simulated users shows 95th-percentile response time ≤ 3 seconds and zero request failures. Fail: Response time exceeds 3 seconds or requests error out under the simulated peak load.	Placement drives create short, high-traffic windows (e.g., the last hour before a deadline); the platform must stay responsive so no eligible student is blocked from applying.

2. UML Use-Case Diagram



3. Use-Case Flow Specification

Use Case ID / Name : UC-02 - Apply to Placement Drive

Actors : Student Applicant (primary) , Resume Parser – system (supporting, via included use case)

Trigger : An eligible Student Applicant chooses to apply to a published corporate drive.

Preconditions :

1. The Student Applicant has an active, verified account.
2. The student's academic profile (CGPA, backlog count, branch) is complete.
3. The target drive is published and its application window is open.

Postconditions (Success):

The application is recorded, the resume is parsed and skill-tagged, and the applicant is queued for Round 1 with status "Applied – Round 1 Queued."

Postconditions (Failure):

No application is recorded; the applicant's status remains "Not Applied" / is set to "Not Eligible" with a discrepancy reason on file.

Main Success Scenario:

1. The Student Applicant logs into the placement portal and opens "Active Drives."
2. The student selects a drive; the system displays the role, CTC, and eligibility rules.
3. The student clicks "Apply" and confirms / uploads their resume (PDF).
4. The system invokes «include» UC-06 "Check Eligibility & Match Profile": it parses the resume, extracts skill tags, and validates the student's CGPA, backlog count, and branch against the drive's eligibility rules.
5. UC-06 returns an "eligible" result to UC-02.
6. The system records the application and queues the student for Round 1 of the interview process.
7. The system displays "Application submitted successfully" and updates the dashboard status to "Applied – Round 1 Queued."
8. The use case ends successfully.

Alternate Flow – A1: Applicant Fails Eligibility Check:

A1 branches from the Main Success Scenario at step 4, when UC-06 evaluates the applicant's profile.

- 4a1. UC-06 detects a discrepancy (e.g., CGPA below the drive's threshold, or an active backlog against a rule that disallows one).
- 4a2. The system displays a rejection message listing the specific discrepancy reason(s) (e.g., "CGPA 6.8 is below the required 7.0").
- 4a3. The system sets the application status to "Not Eligible" and does not create a Round 1 queue entry.
- 4a4. The use case ends without a submitted application; the student may re-apply if their profile is later updated and the drive is still open.